Visualization of Film Circulation and Identifying Patterns of Success

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Abstract. The film festival circuit describes the global network of festivals and how films move between them over time. Various papers have previously explored the idea of interpreting film circulation as a network of interconnected nodes. The core problem therefore does not revolve around the question of whether the film festival circuit can be modelled as such, but rather, what constitutes a tool that maximizes the chances of discovering patterns of success. This thesis consequently presents an interactive, web-based, bipartite, force-directed node graph. The force-directed nature of the final prototype lends itself to the core thesis idea that patterns of success cannot simply be defined by discrete numeric values such as gross revenue, festival showcases, or number of awards, but additionally, through consideration of node placement in the graph and surrounding structures, such as clustering and bridging. Users can analyze the graph via a range of encoded film and festival attributes and restrict the visualized subset via an array of numeric and categorical filters including a user-specific composite score while improving visual distinction by coloring nodes based on a selected attribute. Moreover, the tool is designed with user freedom in mind as rather than dictating to the user what constitutes a pattern of success the tool allows the user to decide for themselves what metrics are valuable indicators to them and approach the topic from their own unique perspective. The final visualization prototype serves as a foundation for future work in the realm of film festival visualization.

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1 Introduction

Film festivals play an important role in determining the length of a film's lifecycle and ultimately its success. Festivals provide directors with the opportunity to showcase their films resulting in increased visibility and recognition as well as the chance to receive awards, further boosting the film's cultural significance. The film festival circuit is especially valuable for smaller, independent films such as short films and documentaries. Thus, it is common practice for film makers and directors to analyze the success of previous similar films to determine an optimal release strategy and what festivals to focus their efforts on.

While large and comprehensive data sets containing detailed information on the current state of film festival circulation already exist, such as the dataset by Loist and Samoilova [LS23], the sheer scale of the dataset makes it incredibly difficult to derive any meaningful conclusions by manual inspection alone. While following a single film's trajectory across festivals can yield beneficial results and insight for directors, finding more general patterns of success across thousands of different data points is nearly impossible without specialized visualization tools or algorithms.

Thus, the scientific problem pertinent to this thesis is what constitutes a visualization tool that allows users to discover meaningful patterns of success across the film festival circuit that can help them construct an optimal release strategy for their own film. The core challenge therefore is not simply to visualize thousands of interconnected nodes, but to reveal clusters, trajectories, and structures. Ideally, such a visualization tool would answer questions such as:

- Do groups of festivals exist that are tightly interconnected?
- Do certain festival node paths exist that increase a film's success rate?
- What festivals are important for a film to be showcased in for certain genres, languages, countries of origin, film type etc.
- What notable differences exist between film type in regard to circulation?

Additionally, it is important to define what constitutes patterns of success and what metrics we can measure to discover them. Indicators of a film's success could include the number of awards a film has won over its lifetime, the number of festivals it was showcased in or its gross total revenue i.e. box-office revenue.

2 Definitions

The following terms are used often throughout the paper and are defined as follows.

Film Festival Circuit: Describes the global network of festivals and how films move between festivals within that network i.e. the relationship between films and festivals over time.

Force-directed node-graph: A network visualization in which nodes and their respective connections are depicted on a 2-dimensional plane that uses physics simulations to arrange connected nodes via spring-like attracting and repelling forces resulting in complex emergent structures.

Bipartite Graph: A graph in which nodes are divided into two separate subsets where connections only occur between nodes belonging to different subsets.

Emergent Properties: Patterns or structural properties that naturally emerge from the physical interaction between nodes in a force-directed node-graph.

Node Degree: The number of outgoing or incoming connections of a node in a network.

Centrality: A metric that measures the structural influence a node has within a network based on its position and relation to neighbouring nodes.

Node Clustering: The degree to which nodes tend to cluster together, measuring the separation within a network.

3 Related Work

3.1 Film Festival Circulation

Historically, film research has mostly relied on either individual qualitative case studies or theoretical frameworks as a way to analyze film circulation. However, in recent years, film circulation research has shifted from qualitative research to data driven quantitative research [LS23]. While qualitative research only captures a certain small subset of film circulation, quantitative research allows for a more comprehensive view that captures circulation patterns of thousands of festivals worldwide. Loist has been pioneering the quantitative approach to film circulation analysis with their papers [LS23] and [Loi19] outlining a comprehensive relational database of films and festivals obtained primarily through manual data collection and web scraping. This thesis will primarily be exploring and making use of this database for the visualization tool as it is arguably the most comprehensive and structured collection of film festival circulation data. Since the database was built with the FAIR(ER) principles in mind, a framework that defines guidelines intended to optimize the reuse of data for humans and computers, finding, accessing and integrating the data will be a far simpler task and potential issues such as missing values, ambiguous groupings and inconsistent naming conventions can be circumvented.

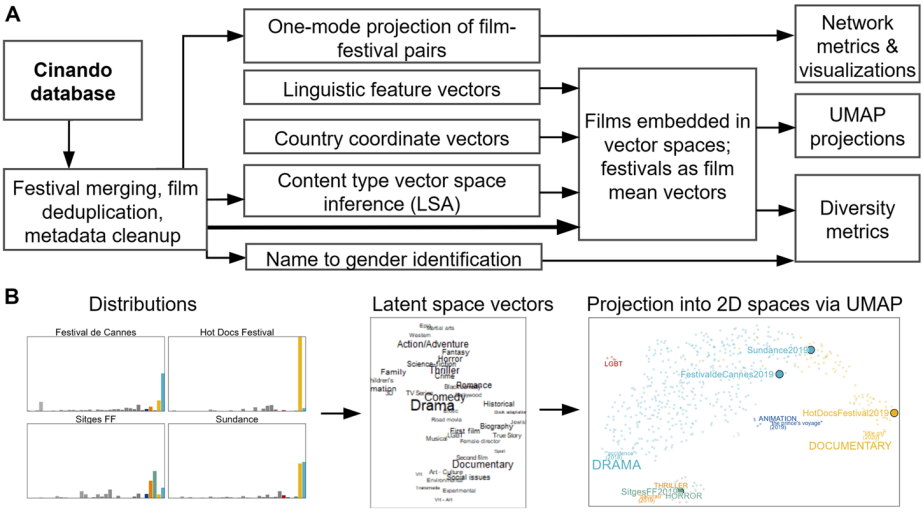


Fig. 1: Quantifying the global film festival circuit: Networks, diversity, and public value creation. [ZKR+24]

Zemaityte et al. [ZKR+24], provide an existing example of a node-based visualization tool that implements a continuous latent vector space system to group festivals instead of discrete labels such as language or production country. While their approach to visualizing the dataset is quite different to the one that will be outlined later in this thesis, it demonstrates that the global film festival circuit can indeed be analyzed as a relational system and reveal meaningful insight,

rather than focusing on isolated events. While Zemaityte et al. [ZKR+24] have already implemented force-directed node graphs for visualizing data sets, the tool itself is primarily used for academic analysis and production of static figures as reference in papers as opposed to an accessible interactive visualization tool for the public use. This thesis will instead focus on a publicly accessible web-based application with interactivity and real-time attribute filtering.

Ehrich et al. [EBSL22] explore a different side to film festival circuit analysis. Specifically, how network analysis can reveal structural inequality in the festival sector. In their paper, they model the sector as a one-mode network of festivals as well as a two-mode network of festivals and films and come to the conclusion that films produced by men-only creative teams are statistically more likely to be showcased at festivals than films by women-only teams.

Loist's article on queer film circulation [Loi15], while mostly qualitative, shows that some films circulate through activist festivals and others through art-film festivals, indicating that a film's circulation is highly dependent on various attributes and it is not prudent to view circulation patterns as homogenous. Additionally, at least back in 2015, although debatably still in today's day and age, society, while portraying itself as having moved into a "post gay" era, still relegates queer centric films to specific LGBTQ activist festivals, while mainstream international art film festivals merely use them as a way of exhibiting artistic or cultural prestige. Loist's paper shows that not only can we discover measurable patterns of success through the visualization of festival circuits but also uncover broader concepts such as societal biases.

3.2 Graph Visualization

Ghoniem et al. [GMM+19] provide a state of the art report on the current research regarding multilayer network visualization, showcasing the differences between multiplex, multimodal, and multilayer representations. Although most of the systems outlined in the paper such as matrix displays and hybrid views exceed the scope of this thesis, it still serves as a valuable resource to determine how a node-based visualization graph should be constructed.

Kobourov's paper [Kob12] on spring embedders and force directed graph drawing algorithms outlines its inherent benefits and drawbacks. Force directed graphs are flexible, aesthetically pleasing and very effective at revealing clusters and groupings, which is perfect for the purposes of discovering patterns of success in film circuits. However, the paper also talks about the scalability issue of calculating physics simulations for a list of growing nodes in such a graph.

Tong et al. [TFGER07] address the issue of finding meaningful sub-structures in large node graphs without relying on exact query matches. They propose a method designed for graphs with nodes containing attributes, which returns not

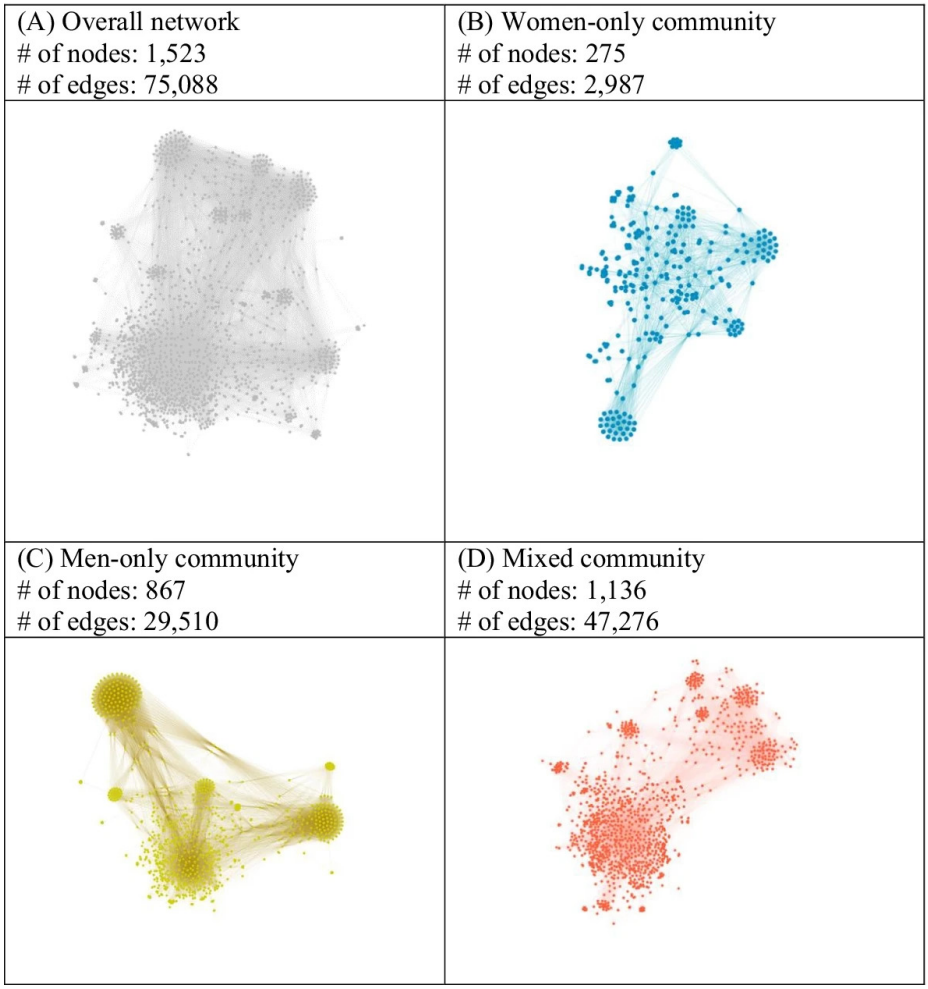


Fig. 2: One-mode film festival network and its gendered communities [EBSL22]

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only exact query matches, but also near matches depending on several parameters. This could serve as a useful feature for discovering circulation patterns in the film festival circuit as some users may be more interested in researching similar circulation patterns to their chosen criteria rather than forcing exact matches, which may inhibit the number of useful sub structures as a result.

Ganser and North [GN98] still provide a foundational overview of the benefits and limitations of using force directed layouts for visualization. The authors argue that force directed layouts specialize in revealing symmetries, structures, and clustering, however, also state that the nodes themselves must be kept simple, as adding in additional labels or descriptions decrease readability.

Holten and van Wijk's paper on force directed edge bundling for graph vi-

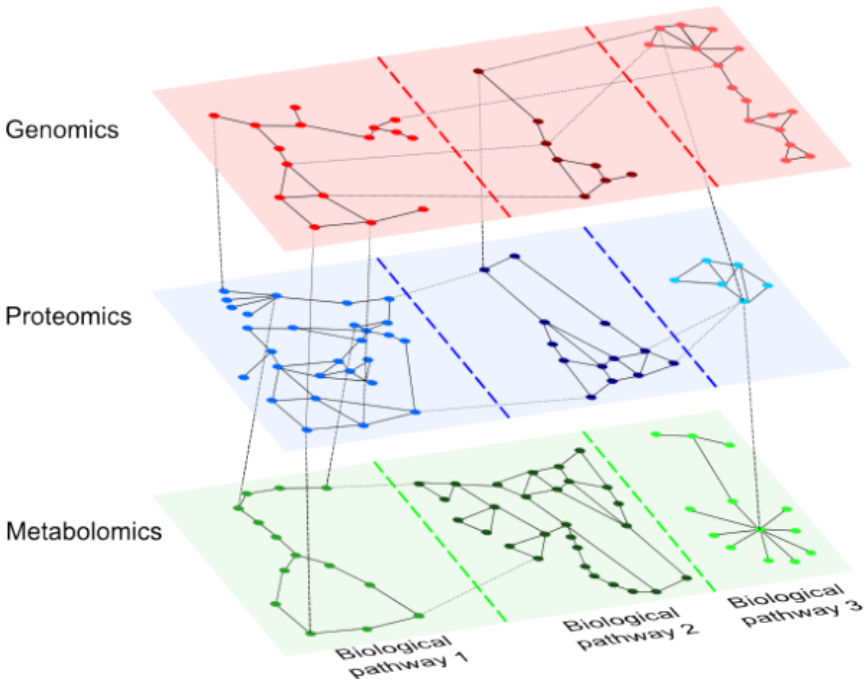


Fig. 3: The State of the Art in Multilayer Network Visualization. A purely illustrative example of multilayer data in the context of biology. The layer can be described by the type of data as a first aspect (genomic, proteomic, or metabolomic), and biological pathway being represented as second aspect. [GMM+19]

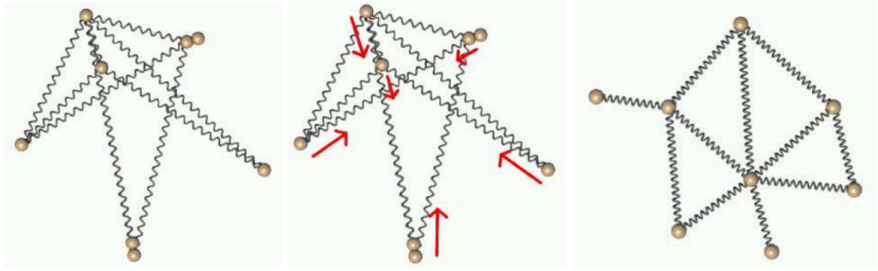


Fig. 4: Illustration of a generic spring embedder: starting from random positions, treat the graph as spring system and look for a stable configuration. [Kob12]

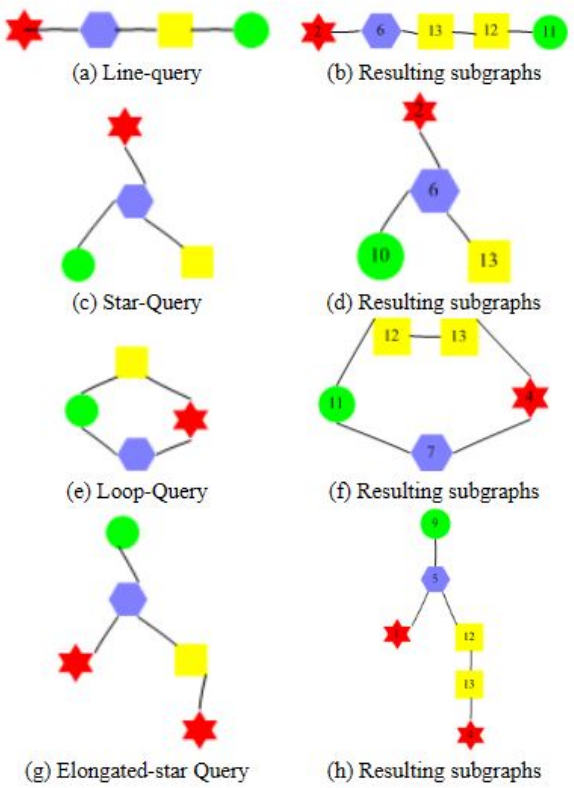


Fig. 5: Examples of queries and results by G-Ray [TFGER07]

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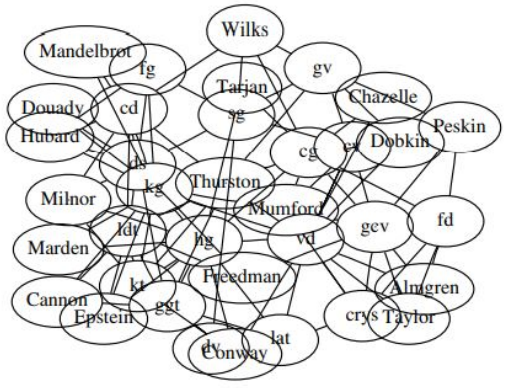


Fig. 6: The effect of non-point nodes [GN98]

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sualization [HvW09] addresses the problem of distinction between edges when nodes become clustered. The paper suggests edge bundling as a way of combining close edges and thus reducing unnecessary clutter while not reducing the graphs usefulness.

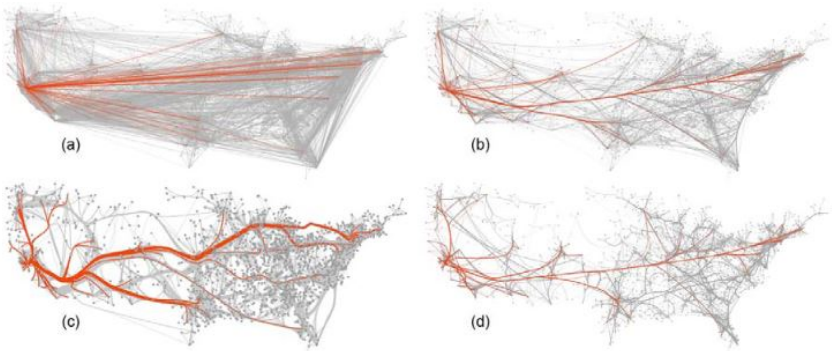


Fig. 7: US migration graph (1715 nodes, 9780 edges) (a) not bundled and bundled using (b) FDEB with inverse-linearmodel, (c) GBEB, and (d) FDEB with inverse-quadratic model [HvW09]

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4 Method

The literature presented in the previous chapter was compiled around the core thesis topic of finding patterns of success in the relational film circulation database. The purpose of the literature review was to assemble a set of papers that together explain the two main ideas surrounding this thesis, namely the film festival circuit and finding patterns of success, and the visualization techniques i.e. primarily bipartite force directed node graphs.

The search strategy revolved around finding relatively new papers, using keywords such as “Film Festival Circulation”, “Finding Patterns of Success in the Film Festival Circuit”, “Bipartite Force Directed Node Graphs”, ”Emergent Properties in Force-Directed Node-Graphs”, and ”Interactive Network Visualization”. Additionally, the papers had to contribute a unique viewpoint or technique as to not have duplicate papers with similar information. Furthermore, in regard to film circulation studies, the methodology had to be mainly quantitative as opposed to qualitative. Regarding the graph visualization portion of the papers, they had to mostly discuss the general framework and methodology of bipartite force directed node graphs, as the final visualization tool will be built using React and the D3.js library as opposed to programming one completely from scratch. Therefore, more math oriented papers were not considered for the purposes of this thesis. Most of the papers were found in open source journals and repositories such as Springer, NECSUS, arXiv, PLOS ONE, Applied Network Science, and ResearchGate.

As aforementioned, the resulting set of papers was compiled as to present a broader view of the thesis statement and topic, as opposed to going into depth into a certain area.

5 Results

The above mentioned literature points to the idea of film circulation being best modeled as a bipartite network of nodes, each containing a list of attributes. This thesis will therefore focus on building a visualization tool that combines the emergent properties of a relational graph view with selective filtering features to limit the complexity of the resulting graph. To emphasize, the data must first be modeled as a network of films and festivals, second, the visualization needs to expose emergent properties such as clusters, bridges, and prominent nodes, and third, it must allow users to filter down the resulting set of nodes by attributes, so as to not overwhelm them with a large, interconnected web of thousands of nodes and links.

5.1 Quantitative Film Circulation Analysis

Loist and Samoilova [LS23] and Zemaityte et al. [ZKR+24] make up the basis of this thesis as, for one, they show that the film festival circuit can indeed be analyzed through quantitative means, as well as visualized using bipartite force-directed node graphs. An ideal visualization tool must model festivals and films as nodes containing filterable attributes.

5.2 Patterns of Success and Emergent Properties

The literature suggests several ways of characterizing patterns of success. One option is to consider the number of festivals a film was showcased in as this indicates widespread appeal across critics and fans. Another way would be to measure the prestige of a film, for example the number of awards it has won or the exclusivity of the festivals it was showcased in. Additionally, considering the gross revenue of a film is a simple yet powerful metric to evaluate success. However, Loist [Loi15] argues that the film festival circuit differs substantially to more commercial oriented releases and cannot be primarily measured using the same metrics of financial achievement. Furthermore, some films may not have been financially successful or have not been shown at a large number of festivals but could still be considered successes due to being influential in their respective sub-circle.

Another important factor to consider when evaluating patterns of success is emergent properties. This includes node clustering, link abundance, and bridges. Metrics that, rather than measuring a single defined numeric parameter, reveal themselves through the physical interaction between nodes and links. The influence of a film may not be strictly tied to discrete values such as prominence or revenue, but also to its influence as a node in such a force-directed graph. For example, a node that bridges two mostly separate portions of a graph may very well be categorized as influential, while not exhibiting the common attributes of an influential film. This supports the idea that allowing users to toggle between various different metrics of success depending on their needs is the optimal path

to achieving a flexible visualization tool, aside from simply discarding the idea of discrete attributes and metrics and instead implementing a latent vector space system as outlined by Zemaityte et al. [ZKR+24].

UnCheol Lee and George A Mashour argue that there are 5 basic network properties i.e. emergent properties that one should be aware of when analyzing force-directed node-graphs [LAM18]. Firstly, the degree property, which simply describes the number of edges attached to a given node. Second, the clustering coefficient, which measures the degree to which nodes tend to cluster together in the graph. Third, centrality, which identifies the most influential nodes within a network. Nodes with a high degree of centrality often bridge two separate parts of a graph, meaning any path connecting these subsets must go through said node. Fourth, path length, which describes the average shortest path distance throughout a network. And lastly, modularity, which reflects the strength of division of a given network i.e. how easily a network can be split up into separate modules.

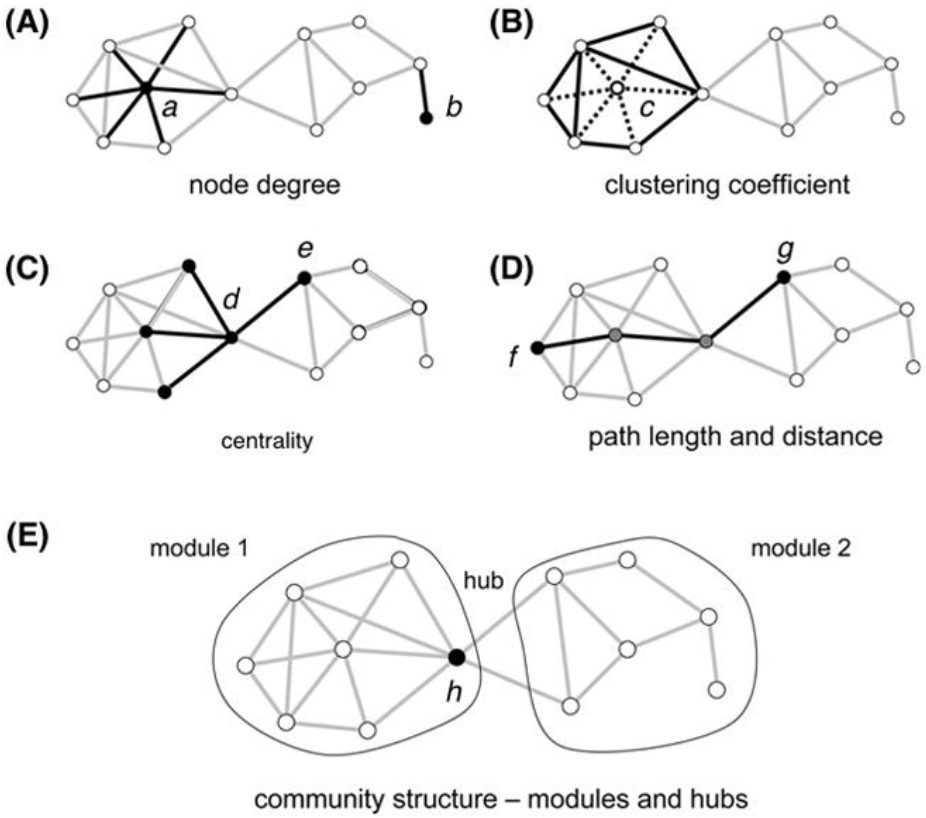


Fig. 8: Illustrations of basic network properties i.e. emergent properties. [LAM18]

5.3 Visualization Models

The literature does not present a single definite best way of modelling the film festival circuit. Most papers implement a node link graph, however, there is no clear consensus on whether using discrete attributes, programming complex layered views and hybrid systems [GMM+19], or discarding the idea of attributes entirely and instead relying on latent vector space [ZKR+24] is the best option. For the purposes of this thesis a simple bipartite force directed node graph will suffice as a prototype. The important aspect is implementing the force-directed layout, as this allows for users to discover emergent properties, which, as previously mentioned, can reveal important patterns of success.

Additionally, Kobourov [Kob12] makes it clear that a well structured filtering system is crucial as to not overload the user with redundant or irrelevant information and to allow the user to focus on the precise metrics they require.

Furthermore, Ghoniem et al. [GMM+19] and Ying et al. [TYLF15] explore how multimodality can be used to create layers of information as to further separate information and make it more manageable for the user. While this is beyond the scope of this thesis, it may be an interesting feature to explore in future research.

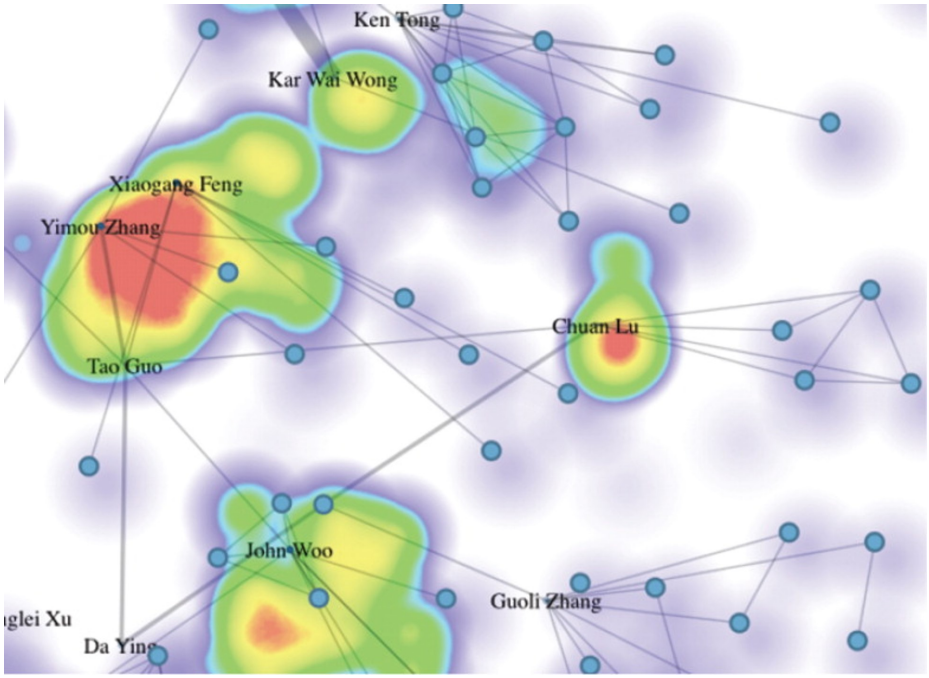


Fig. 9: Visual Analysis of Multimodal Movie Network Data Based on the Double-Layered View. The double-layered visualization combining the node-link view and the density map view. [TYLF15]

5.4 The Database

For the purposes of this thesis the visualization tool will be powered by the film circulation dataset compiled by Skadi Loist and Evgenia Samoilova [LS23]. The database is comprised of four core components: the film dataset containing a set of over 9,000 unique films, a survey dataset covering survey questions, an IMDb dataset comprising awards, companies, crew, festival runs, and film metadata, and a festival dataset which outlines information such as festival name, location, award recognition, and specialization.

The database is incredibly comprehensive and must thus be cleaned, i.e., pre-processed before integrating it into the web application. Many variables appear redundant, relevant in only extremely niche circumstances, or are not useful for discovering patterns of success. While the final set of attributes included in the visualization tool is not final, the following table shows likely candidates for the most important attributes to consider.

Table 1: Dataset components and valuable attributes.

Dataset	**Attributes**
Film	unique.id
	sample.year
	prod.year
	length.min
	prod.country
	regions
	genre
Survey	Screenings
	Budget
	Marketing Costs
IMDb	imdb_awards
	imdb_crew
	imdb_festival_runs
	imdb_release_info
Festival	festival.id
	fest.name
	fest.location
	festival.categories

Additionally, unique identifiers and foreign keys will be useful to draw relations between films, festivals, and their respective surveys and IMDb pages. Due to the sheer number of attributes included in the dataset, in order to achieve a more user-oriented interface, only certain "high value" attributes should be exposed to the user. Hidden attributes may be toggled via an advanced mode settings for those who wish to explore the dataset in more detail.

6 Discussion

The existing literature already showcases that presenting the film festival circuit as a series of interconnected nodes is not only possible, but, generally considered to be the most practical way of discovering valuable metrics and emergent properties, i.e., patterns of success. The thesis will therefore instead focus on how such a visualization tool can be optimized. What does an efficient data to graph view pipeline look like? What features can be implemented to prevent information overload for the user and provide filtering capabilities to focus on certain attributes or sub-structures? What constitutes a pattern of success and how can the tool be optimized to reveal them?

One major advantage of the proposed single layer graph view is that, while perhaps more limited than the latent vector space graph proposed by Zemaityte et al. [ZKR+24], it is far more intuitive which is crucial for a user oriented visualization tool. Users can easily view the dataset in its entirety to discover emergent properties, then filter down the result to inspect clusters and sub-structures in more detail. Additionally, the proposed tool updates its view in real time, as opposed to some of the other models discussed earlier in the literature which are primarily static figure generators.

However, there are some important factors to consider that if left ignored could impact the effectiveness of the visualization tool. For one, if the dataset is too large, even with attribute filtering it may be difficult for users to narrow down their search and discover emergent properties. Additionally, since the tool relies on discrete values in the form of attributes as opposed to a more fuzzy logic oriented approach as previously mentioned in Zemaityte et al.'s paper [ZKR+24], some emergent properties that do not share explicit discrete value similarities may be lost. Furthermore, since the proposed graph view is real time and interactive, depending on the size of the dataset, there may be performance and responsiveness issues on the user end.

To emphasize, the existing literature has gaps when it comes to publicly deployed, interactive, user-oriented visualization tools. This gap is precisely what the final prototype will attempt to fill.

7 Design and Implementation

The final prototype was built around the core thesis idea of, instead of dictating to the user what constitutes a successful film, it is more sensible to provide a large array of filters such that the user may decide for themselves what metrics are valuable to them. The specific implementation was further refined during a meeting with Dr. Loist who, as previously mentioned, is a domain expert in regard to film circulation [Loi19]. The following outlines the features, capabilities, and constraints of the final visualization tool of which the source code is publicliy available on Github [Her26]. Alternatively the tool is deployed live at `ffc.herger.dev`.

7.1 Node Graph

The primary layout is comprised of 2-3 columns depending on whether or not the user is currently inspecting a certain node via the tooltip. The center column contains the node-graph, which is built dynamically, taking every filter setting into consideration when the user clicks on the simulation button, which is located at the top of the right column.

The node-graph was developed using the force-directed node-graph template provided by D3 [D3.23a]. Film nodes are colored blue whereas Festival nodes are colored red. Nodes are connected via links which encode whether a film has been showcased at a certain festival. Every node encodes specific information derived from the database including node type (film or festival), title, category, genre, country, language, year, number of connections, directors, rating, gross revenue etc... For a full list of attributes see 2. While most of the node graph backend is hardcoded, users may use the sliders located in the right column to adjust the minimum and maximum node size, as well as the attraction and repelling forces used when simulating the node-graph.

7.2 Tooltip

If a user wishes to inspect a certain node in more detail they may click the corresponding node and receive detailed information via the tooltip column on the left side of the screen. Both film nodes as well as festival nodes have dedicated tooltips, however, the number of attributes contained within the festival tooltip is less than that of film nodes. Additionally, when a node is selected, all outgoing links will appear yellow, such that the user may easily identify which nodes are connected to the selected node. If a certain attribute is missing from the database, the tooltip will simply display *NA*.

7.3 Categorical Filters

Categorical filters are one of the primary ways users may filter down the node-graph into a smaller subset of nodes depending on their criteria. These exist

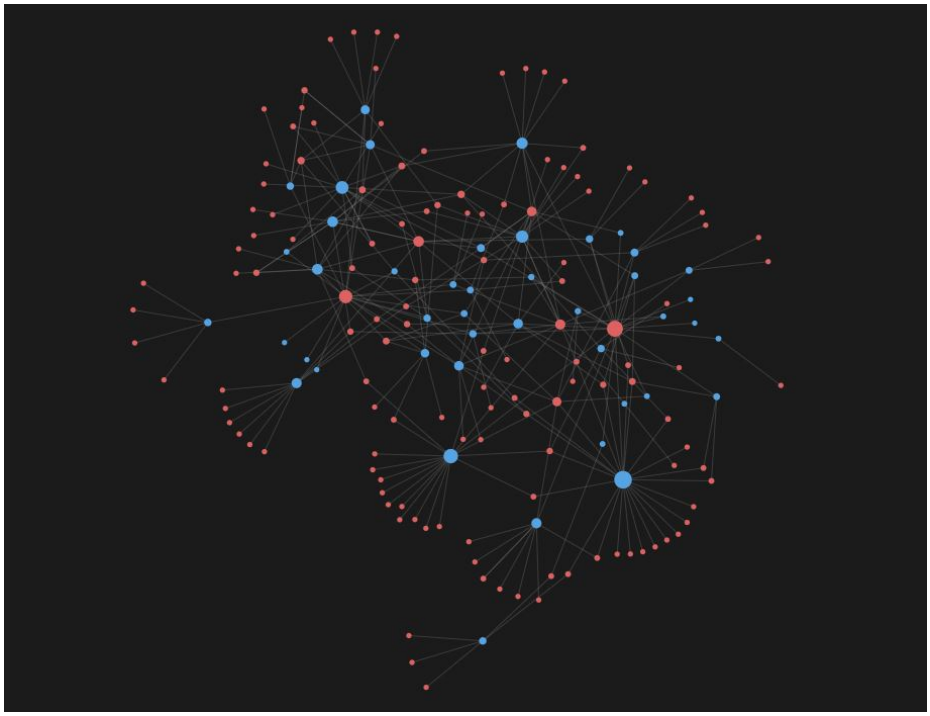


Fig. 10: Example of a filtered-down node-graph simulation.

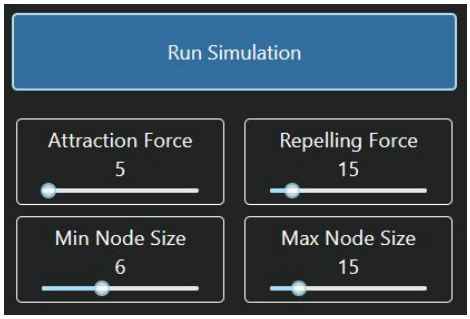


Fig. 11: Force sliders and simulation button.

either as simple checkboxes as is the case with film type and region, or as drop-down selections for attributes containing a large number of unique values such as production country, language, awards etc...

Additionally, below each categorical filter is an *ANDOR* button which users may use to determine whether the filters operate under *AND* logic or *OR* logic.

7.4 Numeric Filters

The second type of filters are numerical including attributes such as production year, film runtime, rating etc... Each numerical filter utilizes a range slider to



Fig. 12: Example of a tooltip.

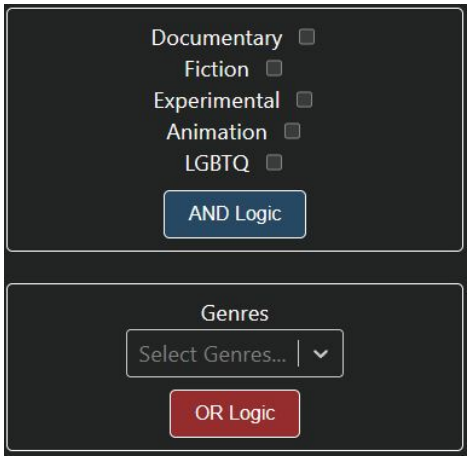


Fig. 13: Example of checkbox categorical filters and dropdown categorical filters as well as *ANDOR* logic buttons.

determine which value interval nodes must inhabit in order to be included in the simulation. Furthermore, above each range slider lies a distribution graph, displaying how a certain attribute is distributed amongst the nodes. Users may use the range slider to narrow the interval in which nodes are visualized which in turn grays-out any bars in the distribution graph that don't lie within the selected interval. The distribution graph component was developed using the histogram template provided by D3 [D3.23b].

After some testing, the ideal axis scaling for the distribution graphs was determined to be linearly on the x axis and Sqrt on the y axis. This allows for high density value intervals not to completely overpower the less frequent intervals while still indicating which values occupy the highest frequency of nodes.

Finally, below each numeric filter lies a *Include NA Values* checkbox which allows users to determine whether nodes with no corresponding value in the database should be included or excluded in the graph simulation.

7.5 Financial Attributes

The visualization tool contains four distinct *financial* attributes, namely: *Budget, Opening USA, Gross USA, and Gross World*. In regard to the distribution graphs, financial attributes are cut off at *$10.000.000* as, while most films lie within this interval, several outliers otherwise unnecessarily compress the graph and make it less legible. Instead, any film that exceeds this threshhold is placed at the value *$10.000.000+*. Additionally, since the values for these financial attributes are so high and users may want to be very specific about their inputs, instead of a range slider these filters are outfitted with two input boxes for the minimum and maximum values.

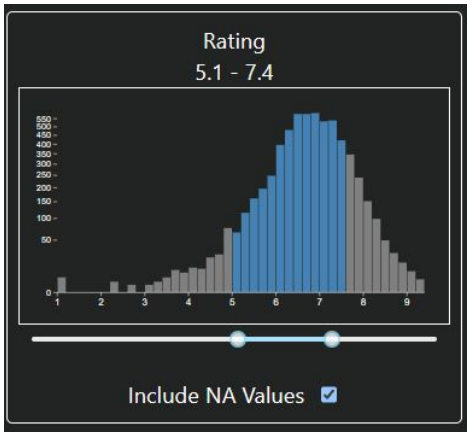


Fig. 14: Example of a numeric filter including a range slider and a corresponding distribution graph, as well as a *Include NA Values* checkbox.

Furthermore, the financial values extracted from the database were widely varied in regard to type of currency. This presented a problem as to how different currencies will be valued when filtering. The developed solution was to create a map of currencies and their respective exchange values compared to the US dollar. Every value was subsequently converted to US dollars as to have a uniform representation across all nodes.

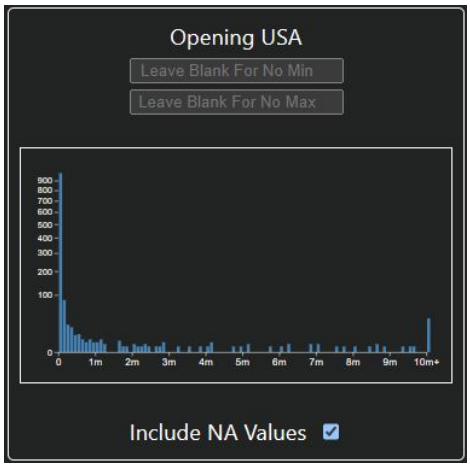


Fig. 15: Example of a financial filter including the clamped distribution graph and the two input boxes for min and max values.

7.6 Festival Filters

The number of festival filters is far fewer when compared to the number of film filters as users will primarily be interested in further film details. The Festival filters include category, country, region, and connections with an additional two

unique filters: *founding year* as opposed to release year, and *city* as films do not possess a release city attribute.

7.7 Composite Score

In the meeting with Dr. Loist, the idea of implementing a composite score feature was brought up, which they agreed could be a very interesting and useful tool for film directors. As such, a composite score has been integrated into the final visualization tool.

By default, the composite score is not activated. Only once the dropdown menu is opened is a score calculated and assigned to each film node. The composite score is essentially a weighted combination of each existing numeric filter. For each numeric attribute users may choose to either maximize, minimize, or optimize for a certain target value, and additionally assign each attribute a weight from 0-100. This allows users to determine which attributes are more or less important according to their own criteria. Each node will subsequently receive a calculated composite score which is visualized via a distribution graph located below the input fields. This can in turn be filtered by the user in order to only simulate nodes whose composite score lies in the chosen interval.

7.8 Node Coloring By Attribute

By default, film nodes are visualized as blue nodes while festivals are depicted as red nodes. However, this does not allow for an easy way to differentiate nodes depending on certain attributes they posess. Thus, a feature to color nodes by their attribute is also included in the final tool.

A dropdown menu allows users to select any existing attribute which will subsequently be applied to the nodes in the simulation. Each node is mapped to a color gradient using D3 depending on its corresponding value. For continuous, numeric attributes this means low values are visualized as red while high values are visualized as blue. For categorical attributes each unique categorical value receives its own designated color, although, for filters with a large number of unique values such as the awards attribute, it becomes increasingly difficult to differentiate values as the number of unique colors diminishes. Furthermore, for categorical attributes, if a node contains multiple values, such as a film that occupies multiple genres, the node will be switched out for a pie chart, depicting the various colors i.e. attributes it posesses.

Festival nodes are colored white in this mode, as black and white are the only remaining colors that can be differentiated from the film nodes mapped to the rainbow gradient.

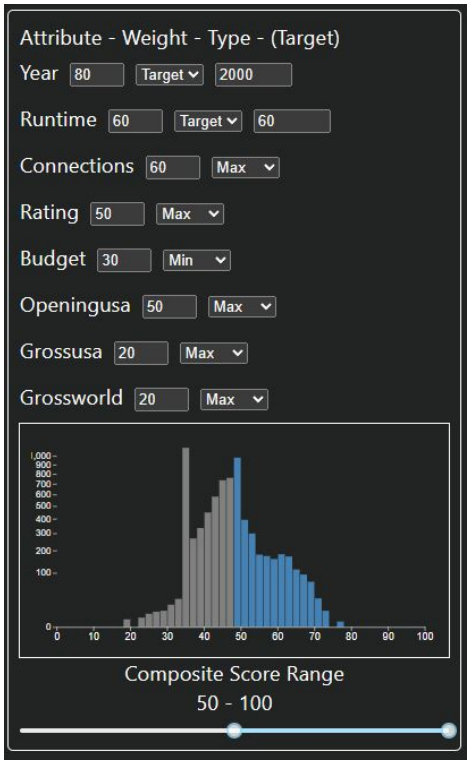


Fig. 16: Composite score dropdown menu including input boxes, type selectors, a range slider, and a distribution graph.

7.9 Emergent Properties

As previously discussed during the literature review portion of this thesis, the ability to discern emergent properties is a crucial component to discovering patterns of success within a force-directed node-graph. However, the bipartite nature of the film festival circulation database presents several limitations to the emergence of typical network structures, and the process of identifying emergent properties differs substantially from conventional node-graphs.

In conventional node-graphs every node in the network may connect to any other node, allowing for the formation of triangles, densely connected neighbourhoods, and communities within the graph. For example, if a certain group of nodes are closely related they may be highly interconnected within the graph and form a dense network with a single node potentially bridging this subset with the rest of the graph. However, with the film festival circulation database, such groups, rather than being tightly interconnected, often have many film nodes connected to a single festival node, where film nodes are instead indirectly connected to each other through a shared festival node. Triangles are simply unable to occur in such a graph due to its bipartite nature, making densely connected neighbourhoods unlikely to occur.

Table 2: Encoded attributes, their type, UI depiction, and node type.

Attribute	**Type**	**UI**	**Node**
Composite Score	Numeric	Input Boxes	Film
Category	Categorical	Checkboxes	Film
Genres	Categorical	Dropdown Selection	Film
Release Type	Categorical	Dropdown Selection	Film
Production Year	Numeric	Range Slider	Film
Runtime	Numeric	Range Slider	Film
Connections	Numeric	Range Slider	Film
Rating	Numeric	Range Slider	Film
Production Country	Categorical	Dropdown Selection	Film
Region	Categorical	Checkboxes	Film
Languages	Categorical	Dropdown Selection	Film
Directors	Categorical	Dropdown Selection	Film
Awards	Categorical	Dropdown Selection	Film
Budget	Numeric	Input Boxes	Film
Opening USA	Numeric	Input Boxes	Film
Gross USA	Numeric	Input Boxes	Film
Gross World	Numeric	Input Boxes	Film
Category	Categorical	Input Boxes	Festival
Production Country	Categorical	Dropdown Selection	Festival
City	Categorical	Dropdown Selection	Festival
Region	Categorical	Checkboxes	Festival
Connections	Numeric	Range Slider	Festival
Founding Year	Numeric	Range Slider	Festival

Moreover, centrality and bridges in particular are not only far more difficult to identify but also occur far less frequently in the graph. In a typical network where groups are connected by a small number of node-bridges the force-directed layout easily reveals the separation, however, in the film festival network there are a large number of nodes connecting very small groups and, even with a high degree of centrality, the node position is highly influenced by its connections. This causes the node to simply appear as a highly connected hub rather than a bridge visually separating distinct communities in the graph.

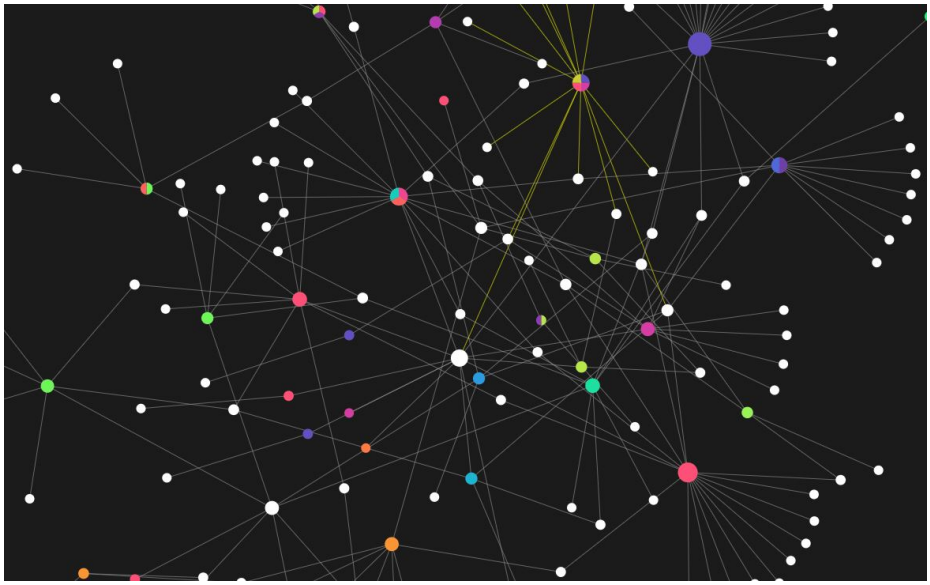


Fig. 17: Example of a graph simulation with nodes colored by their production country.

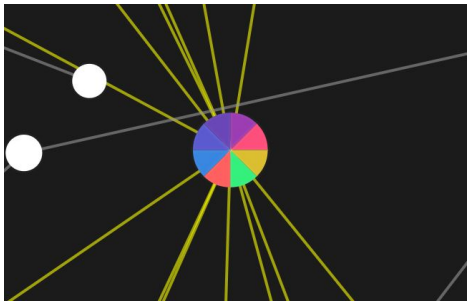


Fig. 18: Zoom in on a node with a large number of unique values depicted as a pie chart.

As such, node degree is the most easily visualized emergent property present within the graph and directly influences the node-size, with centrality and clustering being far more difficult to discern from the layout alone. However, that is not to say that centrality and clustering are absent from the network or impossible to detect, but rather that their visual emergence is drastically constrained by the underlying dataset as the film festival circulation database simply does not produce the same highly connected and separated communities that are typical in denser networks. Thus, users must be aware of the limitations and bipartite nature of the underlying database.

One may assume that due to the bipartite nature of the underlying database, the graph layout should reflect this by instead implementing a bipartite layout, placing nodes along two parallel axes with links between them. However,

according to Al-Naami et al., a user study shows that participants performed significantly worse at discovering node clusters within a network when using a bipartite layout compared to a force-directed layout [ANMMG25]. As such, the regular force-directed layout is still the best for the purposes of visualizing the film festival circuit, even if the discovery of emergent properties is somewhat limited by the nature of the underlying dataset.

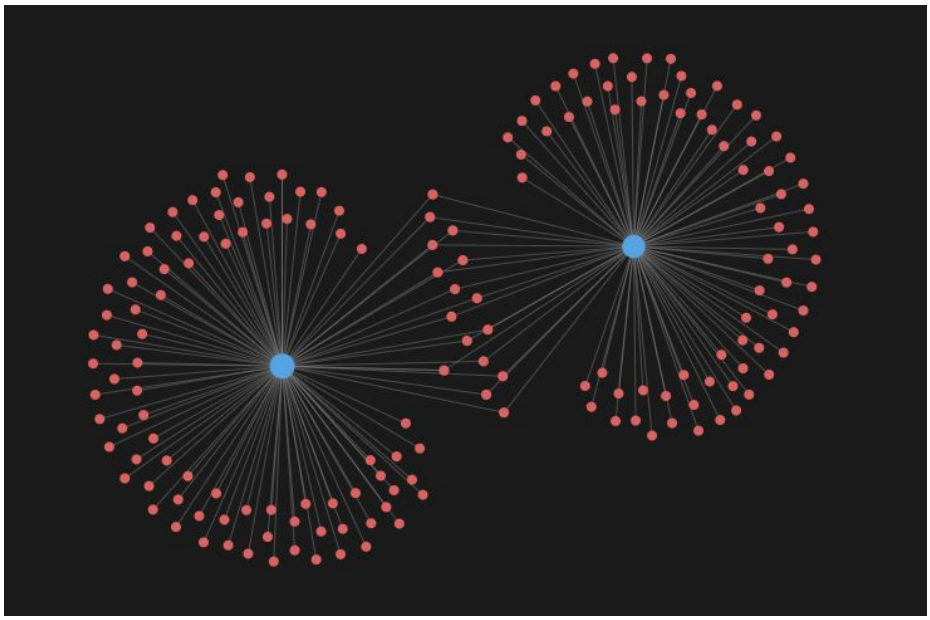


Fig. 19: Film nodes with high degrees within the graph.

7.10 Capabilities and Limitations

Overall, the visualization tool provides a unique contribution and perspective to the realm of film festival circulation. As previously mentioned, the majority of tools that attempt to visualize the film festival circuit operate as a static figure generation tool as opposed to a real-time, interactive, deployed piece of software. This drastically increases the ease of use for directors, and decreases the barrier to entry.

Additionally, being developed around the premise of allowing users to choose for themselves what metrics they consider valuable, the tool provides a high degree of customizability, including the many numeric and categorical filters, but also the ability to alter the physical forces of node interactions via the force range-sliders.

Furthermore, the addition of a composite score is rather unique amongst the existing literature. The abiltity for users to create a custom score using weights for various node attributes further emphasizes the core thesis idea of providing

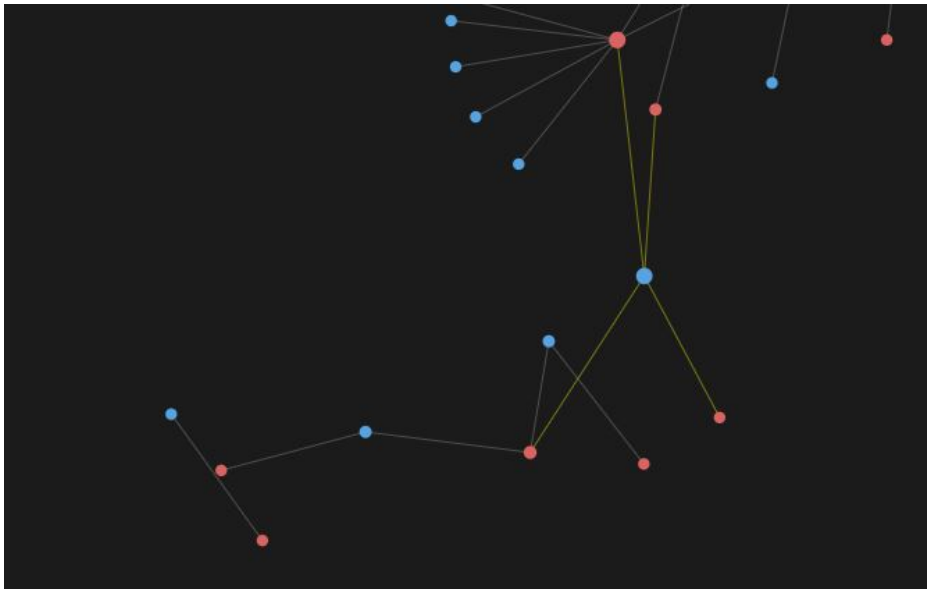


Fig. 20: Film node with a high degree of centrality within the graph, bridging two separate subsets of the network.

users with as many tools as possible without dictating to them what metrics they should emphasize in their research.

However, that is not to say that the tool is devoid of limitations. The fact that the tool was built with real-time interactivity in mind also means that optimizing it will be far more difficult compared to a simple static figure generator. The current database alone, without applying any filters whatsoever, introduces significant performance issues and frame drops due to thousands of nodes and tens of thousands of node-edges being visualized. This is precisely the reason why a simulation button was introduced as opposed to having the simulation automatically run on page load. In this way, users may apply an initial set of filters to narrow down the subset of nodes to something more manageable.

Additionally, many films in the database lack certain attributes and therefore must be depicted as *NA* values. This introduces a dilemma when filtering nodes: When specifiying a certain film category or numeric attribute interval, should films without any value at all i.e *NA* values be included in the resulting subset? Currently this issue is resolved by adding a *Include NA Values* checkbox under every numeric filter such that the user can determine whether or not to keep said values.

Moreover, these *NA* values also affect how the distribution graphs are visualized. Since *NA* values by definition cannot be placed on a specific point on the x-axis, currently, they are simply ignored when building the distribution graphs. This may introduce some bias into the distribution. For instance, it could be the

case that older films are more likely to lack a *year* value due to less strict documentation processes and thus are underrepresented in the depicted distribution graphs.

Another potential issue is the axis scaling. Currently, the x-axis is scaled linearly while the y-axis is scaled by Sqrt where smaller values receive more axis space while larger values are more compressed. This drastically increases the legibility of the graph as for many attributes films tend to congretate around as specific value with less frequent values barely visible under linear scaling. However, this may also skew the perspective of users if they do not realize the difference in scaling on the y-axis.

Moreover, the distribution graphs also suffer from being lopsided. For several of the attributes films tend to either congregate around the minimum or maximum value with a few outliers stretching out the length of the graph, skewing the final visualization to either side. This is especially apparent when it comes to the 4 financial attributes. The vast majority of films tend to have budgets and gross revenues under *$1.000.000* while the remaining films lie somewhere under *$10.000.000*. However, a few individual outliers posess revenues of over *$100.000.000*, dramatically compressing the distribution graph to the left side and making it virtually illegible. The current solution to this is simply to clamp the values between *$0* and *$10.000.000* and introducing a *$10.000.000+* indicator at the end of the graph.

Lastly, the database provides the films budget, gross revenue, and opening revenue in various different currencies. This introduces an issue, namely, how does one filter films depending on their budget or revenue when they are presented in such a non-uniform manner. The solution was to create a map of current currency exchange rates and convert the currencies of each film to the US dollar, thus homogenizing the values. While the original currencies of the films are lost during this process, it is necessary in order to be able to filter films consistently.

8 Conclusions and Future Work

This paper summarizes the existing literature on the topic of visualizing the film festival circuit and presents an interactive visualization prototype implementing a force-directed node-graph. Through representing films and festivals as interconnected nodes, the proposed tool allows users to explore the relationships between films and festivals while focusing on user-specific node subsets through utilization of the extensive filtering functionality including numeric and categorical filters as well as personalized composite scores.

A core consideration during development of the visualization tool was the notion that allowing users to determine for themselves what metrics constitute a success in the realm of film circulation is a far more prudent perspective to approach the issue with as opposed to dictating to the user what metrics they should be focusing on. The final prototype therefore prioritizes freedom and exploration, providing the user with a large array of filters and features such that each user may approach the tool from their own unique perspective. While this paper does not present a definitive definition of what constitutes a pattern of success, it instead attempts to provide an environment in which directors can explore the film festival circuit according to their own criteria.

However, there are still several areas which exceed the scope of this paper and thus require further development. Firstly, expanding and improving the underlying database around which the tool is built would yield significant results. While the database by Skadi Loist and Evgenia Samoilova currently contains the most extensive coverage of the film festival circuit, it could still be expanded upon in order to provide a more complete and holistic view of the landscape [LS23]. Additionally, many attribute values are missing from the current database, and as such, the development of a pipeline to fill in these missing values would be a crucial step in improving the completeness of the database.

Another idea for future improvement would be the integration of a temporal component within the visualization tool. As it currently stands, there is no intuitive way to tell in which order a film was shown at a number of festivals which could serve as additional valuable insight to prospective directors.

Moreover, the current workflow is very much constrained to the web-based visualization tool. It may be a good idea to develop a feature which allows users to export current filters, scores, and graph subsets in a readable manner, such as a spreadsheet, for further external inspection or even to load in as save files as to not lose current settings on page reload.

Finally, arguably the most valuable next step would be to conduct a user study for determining the strengths and weaknesses of the current tool iteration. A user study would ideally invite film directors as the target demographic and investigate what features are the most or least useful for exploring the node-

graph and finding patterns of success. A post-study interview could serve to ascertain potential future features the parcicipants would personally wish to be implemented. Such a user-study would likely be an ideal starting point for future development in the realm of film festival circulation visualization.

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